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Studierichting: Informatica
Oefeningenreeks 4A nr. 54.1

$$\int \cos^{-3} x dx = \int \frac{1}{\cos^3 x} dx$$
$$= \int \frac{\sin^2 x + \cos^2 x}{\cos^3 x} dx = \int \left(\frac{\sin^2 x}{\cos^3 x} + \frac{\cos^2 x}{\cos^3 x} \right) dx$$
$$= \int \frac{\sin x \cdot \sin x}{\cos^3 x} dx + \int \frac{1}{\cos x} dx$$
$$= \left(\sin x \cdot \frac{1}{2\cos^2 x} - \int \frac{1}{2\cos^2 x} \cdot \cos x dx \right) + \int \frac{1}{\cos x} dx$$
$$= \frac{\sin x}{2\cos^2 x} - \frac{1}{2} \int \frac{\cos x}{\cos^2 x} dx + \int \frac{\cos x}{\cos^2 x} dx$$
$$= \frac{\sin x}{2\cos^2 x} - \frac{1}{2} \int \frac{\cos x}{1 - \sin^2 x} dx + \int \frac{\cos x}{1 - \sin^2 x} dx$$
Stel $\sin x = t$ dan $dt = \cos x dx \Leftrightarrow dx = \dfrac{dt}{\cos x}$
$$= \frac{\sin x}{2\cos^2 x} - \frac{1}{2} \int \frac{dt}{1 - t^2} + \int \frac{dt}{1 - t^2}$$
$$= \frac{\sin x}{2\cos^2 x} - \frac{1}{2} \cdot \left(\frac{1}{2} (\ln |1 + t| - |1 - t|) \right) + \frac{1}{2} \cdot (\ln |1 + t| - |1 - t|) + c$$
$$= \frac{\sin x}{2\cos^2 x} - \frac{1}{4} \ln \left| \frac{1 + t}{1 - t} \right| + \frac{1}{2} \ln \left| \frac{1 + t}{1 - t} \right| + c$$
$$= \frac{\sin x}{2\cos^2 x} + \frac{1}{4} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| + c$$

References